

Review

Progress and Limitations in Forest Carbon Stock Estimation Using Remote Sensing Technologies: A Comprehensive Review

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Abstract: Forests play a key role in carbon sequestration and oxygen production. They significantly contribute to carbon peaking and carbon neutrality goals. Accurate estimation of forest carbon stocks is essential for a precise understanding of the carbon sequestration capacity of forest ecosystems. Remote sensing technology, with its wide observational coverage, strong timeliness, and low cost, is essential for carbon stock estimation research. However, challenges in data acquisition and processing include data variability, signal saturation in dense forests, and environmental limitations. These factors hinder accurate carbon stock estimation. This review summarizes the current state of research on forest carbon stock estimation from two aspects, namely remote sensing data and estimation methods, highlighting both the advantages and the limitations of various data sources and models. It also explores technological innovations and cutting-edge research in the field, focusing on deep learning techniques, optical vegetation thickness estimation methods, and the impact of forest–climate interactions on carbon stock estimation. Finally, the review discusses the current challenges in the field, including issues related to remote sensing data quality, model adaptability, forest stand complexity, and uncertainties in the estimation process. Based on these challenges, the paper looks ahead to future trends, proposing potential technological breakthroughs and pathways. The aim of this study is to provide theoretical support and methodological guidance for researchers in related fields.

Keywords: data source synergy; estimation methods; deep learning; AI-driven innovation



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1. Introduction

Since the Industrial Age, the increased use of fossil fuels has led to a significant rise in greenhouse gas emissions, particularly carbon dioxide (CO₂), a key driver of global warming [1]. This disruption to the ecological balance poses a threat to human living conditions. In 2024, global CO₂ emissions are projected to reach 41.6 billion tons, exceeding the 40.6 billion tons recorded in 2023, according to the “Global Carbon Project” in their Global Carbon Budget report [2]. Achieving the Paris Agreement’s target of limiting the global temperature rise to well below 1.5 °C and implementing China’s dual carbon strategy, which aims to peak carbon emissions by 2030 and achieve carbon neutrality by 2060, remain major challenges. Addressing global climate change requires not only reducing carbon

emissions but also enhancing carbon sinks. Strengthening the resilience and stability of ecosystems is crucial to increasing their carbon sequestration capacity [3].

Forests, as the largest and most biologically diverse terrestrial ecosystems, have a unique capacity for carbon sequestration and oxygen production [4]. Through photosynthesis, trees absorb atmospheric CO₂ and store it in forest biomass, making them one of the most effective and economically viable methods to reduce atmospheric CO₂ levels [5]. However, forest carbon sequestration is influenced by forest dynamics and human activities. Forest carbon stocks and the carbon cycle are critical to understanding the role of terrestrial ecosystems in global carbon cycling and climate change [6]. Monitoring carbon stocks and their temporal changes is essential for reducing uncertainties in broader ecosystem carbon cycle models.

Traditional forest inventory methods, which measure tree height, diameter at breast height (DBH), and crown width within sample plots, provide accurate data on forest characteristics. These inventories have been used to develop and validate carbon stock estimation models [7]. For instance, Fang Jingyun et al. [8] utilized fifty years of China's continuous forest inventory data to assess changes in the country's forest carbon stocks from 1949 to 1998. Similarly, Andreas Margerl et al. [9] employed the United States' national, historical, and recent forest inventory data spanning from 1907 to 2012 to evaluate the dynamic variations in carbon content within the native forest ecosystems of the United States during this period. Despite their high accuracy, these traditional methods are time-consuming, labor-intensive, and require tree felling, which can damage ecosystems. Thus, these methods are not scalable or cost-effective for meeting growing demands for forest data.

With the widespread adoption of satellite remote sensing technology and the commercialization of geospatial data in the 1980s, this field has rapidly advanced [10]. However, beyond satellites, other platforms, like aircraft and unmanned aerial vehicles (UAVs), have also emerged as powerful tools in forest monitoring. Aircraft-based observational platforms provide higher spatial resolution and more flexible observation schedules compared to satellites, making them particularly useful for monitoring forest dynamics on a regional scale. UAVs, with their ability to fly at low altitudes, enable highly detailed measurements of forest structure and biomass, offering a cost-effective solution for local-scale monitoring and dynamic changes in forest carbon stocks. These platforms complement satellite data by filling gaps in spatial and temporal resolution, providing more accurate and timely estimates of carbon stocks [11]. Compared to traditional ground-based inventories, which are labor-intensive, time-consuming, and require tree felling, remote sensing platforms provide a more efficient and less invasive method for assessing forest structure and carbon content. Figure 1 illustrates this comparison, highlighting the strengths and limitations of both traditional and remote sensing methods in mapping forest carbon storage distribution.

To comprehensively evaluate the estimation methodologies of carbon stocks utilizing observational technology, an extensive literature search was conducted on 10 December 2024 within the Web of Science Core Collection database. The search employed the keywords "forest carbon stocks" and "remote sensing". A total of 2191 publications, encompassing scientific research articles and other relevant online media literature published between 1999 and 2024, were systematically reviewed and analyzed. The findings were subsequently synthesized, highlighting the annual publication trends, the geographical distribution of research outputs, and the distribution across various academic journals, as illustrated in Figure 2.

Figure 2a illustrates a significant increase in both the number of studies and article citations concerning the estimation of forest carbon stocks using remote sensing technology from 2000 to 2024. Notably, since 2022, the annual publication count has exceeded 400 pa-

pers for three consecutive years, and the cumulative citation count has surpassed 15,000. This trend reflects the heightened global research activities related to forest above-ground biomass (AGB), driven by the emergence of environmental protection initiatives, such as the Paris Agreement and China's commitments to carbon peak and carbon neutrality. Figure 2b depicts the distribution of publications based on the country of the first author's institutional affiliation. The United States, China, and Germany are the leading contributors, with 1126, 1079, and 269 papers, respectively. Figure 2c presents the top 10 journals ranked by publication volume in this domain, highlighting highly esteemed journals in the field of remote sensing, including *Remote Sensing* (690 papers) and *Remote Sensing of Environment* (344 papers). These journals are recognized as primary platforms for disseminating research on forest carbon stock estimation and related remote sensing applications. These trends clearly demonstrate the critical role of remote sensing technology in advancing forest carbon monitoring.

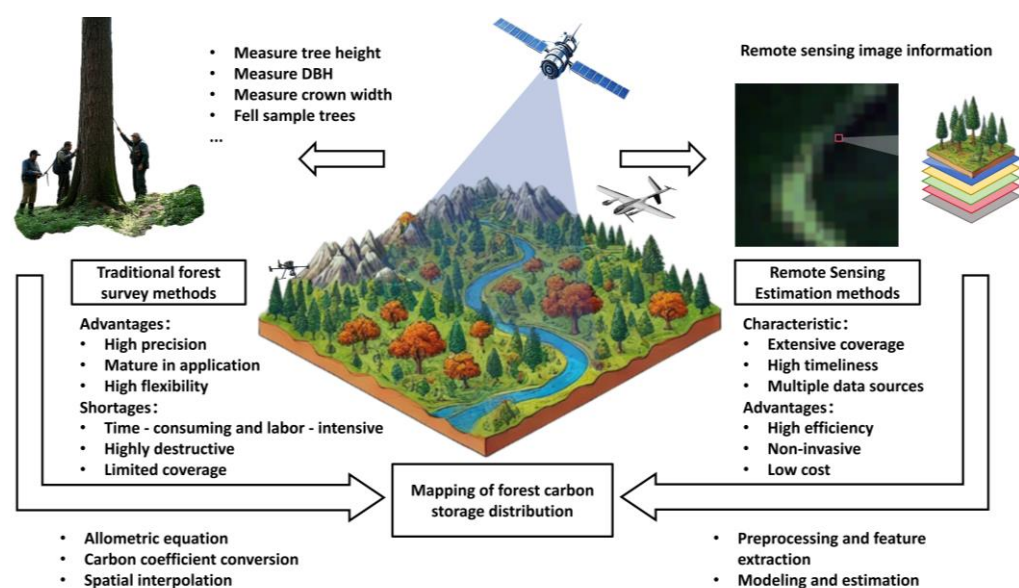


Figure 1. Comparison of traditional forest survey methods and remote sensing estimation methods for forest carbon storage mapping.

With the advent of the era of big geospatial data and the advancement of sensor technologies, the volume of remote sensing data applicable to the estimation of forest carbon stocks is increasing exponentially [12]. Concurrently, developments in computer science have introduced novel methodologies for constructing estimation models [13,14]. Remote sensing data serve as the foundational observational information, while estimation techniques are employed to analyze and infer carbon stocks based on this data.

This paper aims to fill gaps in the current literature by providing a comprehensive review of remote sensing-based forest carbon storage estimation, focusing on technological advances and the limitations inherent in current methodologies. Furthermore, the paper will discuss how current methods can be better applied in real-world contexts, such as climate policy formulation and forest management practices, where precise carbon stock data are crucial for informed decision-making. By presenting an in-depth overview of these areas, this study seeks to supplement existing knowledge and provide insights into future research directions. The ultimate goal is to offer a more nuanced understanding of the challenges in remote sensing-based carbon stock estimation and to propose potential solutions that could advance both theoretical and practical applications of this field.

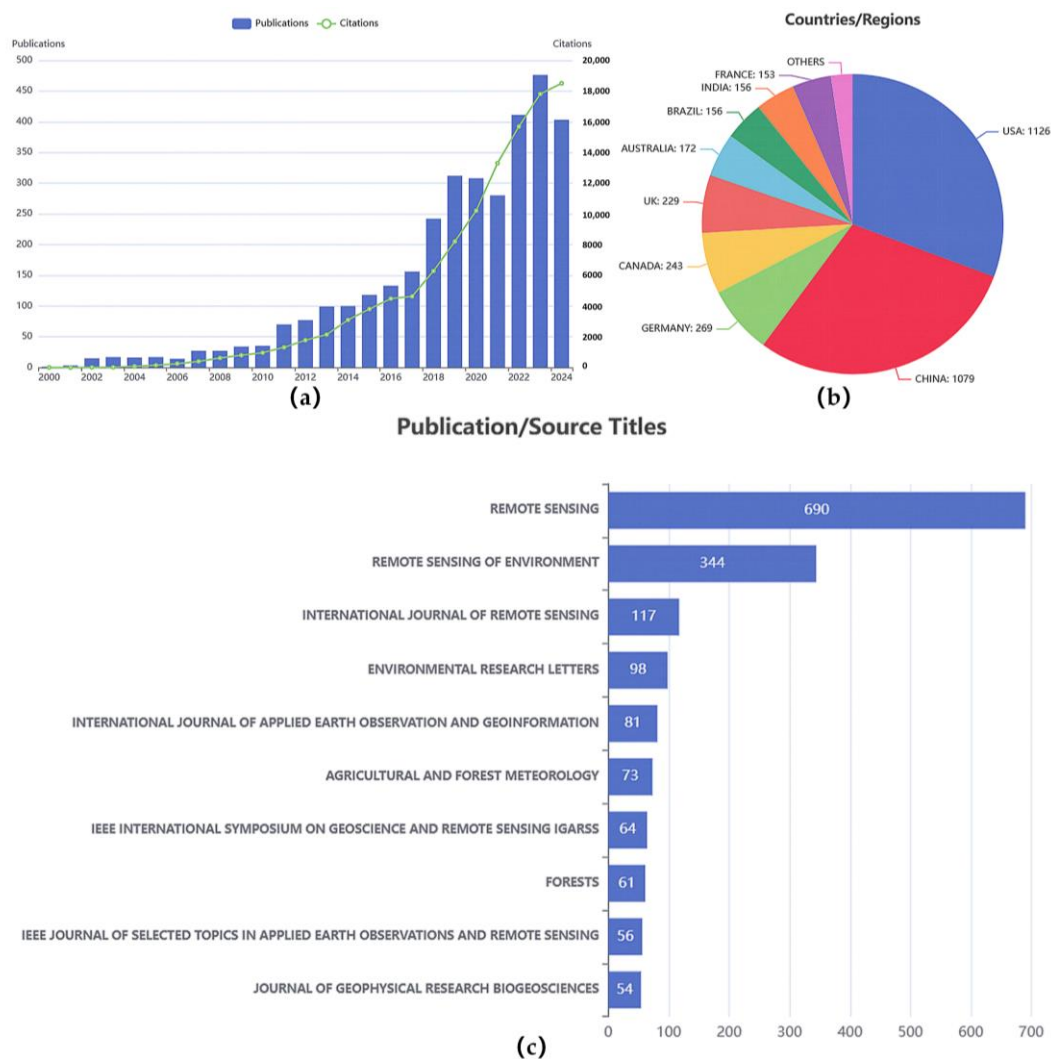


Figure 2. Trends, geographic distribution of the authors or research institutions, and source titles of publications on forest carbon stock estimation using remote sensing. (a) Annual number of publications and citations from 2000 to 2024, showing a significant increase in both over time. (b) Distribution of publications by country/region, highlighting major contributing countries such as the USA and China. (c) Top publication sources, listing the most frequently appearing journals in this research field, with “Remote Sensing” leading the count.

2. Remote Sensing Data Sources

Remote sensing data can be categorized into optical, synthetic aperture radar (SAR), and light detection and ranging (LiDAR) data based on the sensor type used [15]. These datasets, with varying spatial resolutions and spectral bands, support the estimation of forest carbon stocks.

2.1. Optical Data

Optical remote sensing data are crucial for estimating forest carbon stocks due to variations in reflectance spectral curves across plant species in the visible and near-infrared (NIR) bands. Even within a single species, reflectance can fluctuate based on environmental conditions and phenological stages [16]. These spectral characteristics enable continuous monitoring of plant growth and accurate carbon stock estimation, establishing optical data as a widely used modality in forest carbon stock research. Common reflectance indices, such as the normalized difference vegetation index (NDVI), are used to assess vegetation health and biomass [17], while the enhanced vegetation index (EVI) improves upon the

NDVI by reducing atmospheric and soil noise. The photochemical reflectance index (PRI) offers insights into plant photosynthetic activity, further enhancing carbon stock estimation accuracy. Optical data are primarily categorized into multispectral and hyperspectral data.

2.1.1. Multispectral Data

The use of optical remote sensing for carbon stock estimation began in the 1970s, initially with single-band imagery. However, the accuracy of early methods was impacted by atmospheric conditions, soil characteristics, and sensor performance. The use of multispectral data, combining multiple spectral bands, improved accuracy by better correlating with forest biomass and carbon stocks [18]. Today, multispectral datasets, such as Landsat, WorldView, and the GF series, are widely used in forest carbon stock estimation. These datasets are commonly processed using regression techniques that model relationships between carbon stocks and spectral, terrain, and texture variables.

Advancements in sensor technology have led to higher spatial resolution imagery, which aids in classification tasks, such as tree species identification, improving carbon stock estimations [19]. However, high-resolution imagery often lacks comprehensive spectral information, and its higher cost and susceptibility to weather conditions present challenges in practical applications [20].

2.1.2. Hyperspectral Data

Hyperspectral data capture biophysical and chemical characteristics more precisely than multispectral data, making them valuable for carbon stock estimation. Hyperspectral technology originated from the imaging spectrometer research program in the United States. It was first proposed by researchers at the Jet Propulsion Laboratory of the California Institute of Technology and has since made significant advancements with the support of NASA [21]. Compared to multispectral data, hyperspectral data offer the capability to capture biophysical and chemical characteristics across various temporal and spatial scales, utilizing finer spectral resolution bands and providing more comprehensive spectral information [22]. Additionally, the utilization of narrow-band vegetation indices derived from hyperspectral data can mitigate the effects of atmospheric and water absorption, as well as address the saturation issues associated with broadband vegetation indices [23].

However, the large number of spectral bands in hyperspectral data can lead to redundancy, increasing processing and storage costs. Additionally, hyperspectral sensors are more susceptible to atmospheric conditions, requiring rigorous correction techniques to ensure accurate data [22]. Despite these challenges, hyperspectral data provide significant potential for enhancing carbon stock estimates.

2.1.3. Progress in Forest Carbon Stock Estimation Based on Optical Data

Optical data were first applied to the estimation of biomass and carbon stocks, extracting vegetation parameters from images, generating values indicative of biomass through band calculations, and using this information to estimate aboveground biomass of forests (Table 1). This approach is widely used because of its simplicity and high estimation accuracy.

However, there are still certain limitations in the application of optical data in the study of forest carbon stock estimation. Firstly, optical images may experience a decline in image quality and data loss due to differences in sunlight and cloud interference. Secondly, optical data mainly acquire spectral information within the ground area covered by the image, only providing information on the horizontal distribution of the forest, but do not offer vertical distribution information. Finally, in areas with high forest canopy density and accumulation, optical data may exhibit oversaturation, resulting in a lower estimation of carbon stocks.

Table 1. Recent studies on forest carbon stock estimation using optical data.

Study	Study Area	Data Source	Highlight	Accuracy	Method
Halme et al. [24]	Hyytiälä Station, Finland (<i>boreal coniferous forest</i>)	AISA: 0.7 m; Sentinel-2: 10 m	Hyperspectral data improve LAI estimates, while Sentinel-2 provides similar accuracy for height, basal area, and biomass.	$R^2 = 0.63$ RMSE = 21.46	SVR, GPR, 120 plots, 5-fold cv
Verly et al. [25]	Federal University of Viçosa, Brazil (<i>montane semi-deciduous forest</i>)	Sentinel-2: 10 m	Vegetation indices did not improve carbon prediction, but rainy-season spectral variables (blue, green, red, NIR) enhanced accuracy.	$R = 0.9871$ RMSE = 4.1696	Artificial neural networks selection, 30 plots
Li et al. [26]	Hunan Province, China (<i>subtropical evergreen broadleaf forest</i>)	Landsat 8 OLI: 30 m	Seasonal Landsat 8 imagery yielded varied forest AGB estimates and highlighted uncertainties in biomass estimation.	$R^2 = 0.55$ RMSE = 19.45	Stepwise regression, 5-fold cv, 303 plots
Sa et al. [27]	Saihanba Forest, Hebei, China (<i>temperate coniferous forest</i>)	Sentinel-2: 10 m; Landsat 8 OLI: 30 m	The new vegetation and texture ratio indices outperform traditional methods in accurately reflecting forest changes and estimating canopy closure, basal area, and AGB.	$R^2 = 0.6991$ RMS = 3.2687	Linear regression, 65 plots

2.2. SAR Data

SAR data offer advantages over optical data, including a broader wavelength range, enhanced penetration, and reduced susceptibility to weather conditions, making them valuable for forest carbon stock estimation [28]. SAR data allow for the acquisition of vertical structural information about forests, which is critical for accurate carbon stock calculations [29].

2.2.1. Classification of SAR Data

SAR data can be classified based on polarization modes, frequency bands, and scattering mechanisms [30], with each offering distinct benefits for carbon stock estimation.

- **Polarization mode:** Single polarization provides basic forest coverage and structural classification. Dual polarization improves vegetation type differentiation and biomass estimation, while full polarization enables precise biomass and carbon stock estimation, particularly in complex forests. Cross-polarization increases sensitivity to volume scattering, beneficial for dense forest structures [31].
- **Frequency bands:** The P-band and L-band are best for estimating biomass and tree trunk data due to their strong penetration capabilities [32]. The C-band and X-band are used for monitoring forest canopy structures [33], while the Ku-band and Ka-band, which are more suited for monitoring forest health, are less effective for direct carbon stock estimation [34].
- **Scattering mechanisms:** Volume scattering is key for evaluating forest density and biomass, while double scattering helps estimate trunk density in densely vegetated areas. The backscattering coefficient, influenced by forest structure, aids in inferring biomass and forest health [35].

By utilizing these different classifications, SAR data provide a comprehensive approach to estimating forest carbon stocks, offering detailed insights into forest structure and biomass.

2.2.2. Estimation of Forest Carbon Stocks Using Multi-Frequency SAR Data Synergy

Recent advancements in SAR technology allow for multi-frequency data integration, enhancing the accuracy of forest carbon stock estimations. Polarimetric interferometric SAR, when combined with interferometric phase data, helps delineate the three-dimensional structure of forests, particularly the vertical distribution of tree heights [36]. These 3D structural data are essential for precise biomass and carbon stock estimation [37].

The use of multi-frequency SAR data also addresses the limitations of optical data, which struggles under rainy or cloudy conditions, enhancing the reliability of forest carbon estimates. Additionally, the robustness of SAR data against atmospheric disturbances allows for continuous, reliable monitoring of forest ecosystems, improving long-term forest management and climate mitigation efforts.

2.2.3. Progress in Forest Carbon Stock Estimation Based on SAR Data

SAR data can provide high-quality surface cover information under various weather conditions and is unaffected by cloud cover and precipitation, making it particularly suitable for estimating forest carbon stocks. Through multi-temporal analysis of SAR data, it is possible to effectively monitor changes in forest structure, improving the accuracy and timeliness of carbon stock estimation. Table 2 summarizes recent studies on forest carbon stock estimation using SAR data.

Table 2. Recent studies on forest carbon stock estimation using SAR data.

Study	Study Area	Data Source	Accuracy	Highlight	Parameters	Verification
Li et al. [38]	Monda test site near the equator	F-SAR	$R^2 = 0.947$	Developed a TomoSAR-based approach to estimate tropical forest AGB by extracting vertical structural parameters, including under-canopy terrain, vertical backscatter power, canopy height, and total canopy backscatter power.	HH, HV, VV	Existing inventory data
Hu et al. [39]	Saihanba Forest, Hebei, China	GF-3, ALOS-PALSAR	$R^2 = 0.93$	Combining C- and L-band data provides a more comprehensive representation of forest canopy information, enhancing the saturation point in biomass estimation.	HH, HV, VH, VV; wavelength = 5.5 cm, incidence angle = 21.3°; wavelength = 23.6 cm, incidence angle = 27.8°	Field data from 50 plots
Ali et al. [40]	Hardevan Forest, India	TanDEM-X, ALOS-PALSAR	$R^2 = 0.704$	Utilized time series L-band backscatter data to assess the impact of different polarizations on the accuracy of AGB and forest height estimates, enabling their joint inversion.	HH, HV, VH, VV	Field data from 60 plots
Ramachandran et al. [41]	Palaku Moist Tropical Forest, French Guiana	TropiSAR	$R^2 = 0.89$	TomoSAR is promising for AGB estimation compared to LiDAR; their combination facilitates regional and global AGB mapping.	HH, HV, VH, VV; wavelength = 0.75 cm, incidence angle = 24–62°	Airborne LiDAR and inventory data

Despite the advantages of SAR images in data acquisition and carbon stock estimation, the associated data processing procedures are complex. SAR data also show limited

sensitivity to vegetation type variations. In areas with complex terrain, such as mountainous regions, radar signals are affected by terrain undulations, leading to terrain-induced deviations in radar echo intensity. These effects can hinder the accurate estimation of forest structure and biomass, especially in steep regions. Furthermore, the backscattering coefficient, commonly used in SAR data applications, can produce inaccurate results due to its sensitivity to changes in forest structure and aboveground biomass. While SAR performs well in estimating carbon stocks in low-density forests, its effectiveness diminishes in high-density forests, where the correlation between radar backscatter and biomass weakens, leading to saturation effects. These challenges highlight the need for advanced processing techniques and integrated approaches to improve the accuracy and reliability of SAR-based forest carbon stock estimations, especially in dense and topographically complex forest ecosystems.

2.3. LiDAR Data

LiDAR is an active remote sensing technology that efficiently captures precise three-dimensional spatial data by emitting laser pulses and detecting their reflected signals [42]. It is particularly advantageous for estimating forest carbon stocks due to its ability to accurately capture vertical forest structure and overcome signal saturation issues typical of optical and SAR data. These capabilities significantly improve biomass and carbon sequestration assessments, providing more reliable and detailed information. LiDAR's high spatial resolution aids in developing advanced models for forest structure analysis, making it a key tool in forest management and climate change mitigation strategies [43].

2.3.1. Airborne LiDAR

Airborne LiDAR, deployed on aircraft or UAVs, offers high-precision three-dimensional measurements of forest ecosystems [44]. It has become an essential tool for estimating forest carbon stocks due to its superior spatial resolution and coverage compared to traditional methods. Airborne LiDAR efficiently surveys inaccessible areas, capturing detailed structural information, such as canopy height, density, and understory terrain. This enables accurate biomass estimation, differentiation of forest types, and analysis of forest gaps and edges, which often have significantly different carbon stock values [45]. Furthermore, UAV technology advancements and reduced laser costs have made airborne LiDAR more accessible, expanding its application in complex, large-scale terrains [46].

2.3.2. Spaceborne LiDAR

Spaceborne LiDAR, mounted on satellites, provides large-scale, high-precision measurements of the Earth's surface, including forest vertical structures [47]. Its ability to cover vast areas from orbit makes it ideal for regional and global carbon stock monitoring. Spaceborne LiDAR can measure parameters, such as canopy height and area, helping to distinguish between forest types and estimate biomass [48]. This technology supports global carbon stock assessments, especially in remote or difficult-to-reach areas, like tropical rainforests [49]. Notable missions, such as NASA's GEDI and ICESat-2, are advancing forest structure data acquisition, refining global carbon cycle models, and enhancing our understanding of forests' role in climate regulation.

2.3.3. Progress in Forest Carbon Stock Estimation Based on LiDAR Data

LiDAR, as an advanced active remote sensing detection method, operates by independently emitting laser pulses toward target ground objects and analyzing the returned signals to obtain precise measurements of stand height and vertical forest structure. This capability distinguishes LiDAR from other remote sensing technologies, enabling the acquisition of detailed three-dimensional information that is unattainable through alter-

native methods. Table 3 presents recent studies on forest carbon stock estimation using LiDAR data.

Table 3. Recent Studies on Forest Carbon Stock Estimation Using LiDAR Data.

Study	Study Area	Data Source	Accuracy	Highlight	LiDAR Parameters	LiDAR-Derived Metrics
Bazezew et al. [50]	Aye Xitan Tropical Rainforest Reserve, Malaysia	ALS, TLS	$R^2 = 0.96$	Integrating ALS and TLS data estimates tropical forest AGB more accurately than traditional field methods, reducing underestimation and improving carbon stock assessments.	Pulse rate = 70 kHz and 240 kHz, scan angle = 60° , laser points/m ² = 5 to 6 points with 808 m to 1155 swath width	CHM
Zhou et al. [51]	Zhejiang A&F University, China	ULS	$R^2 = 0.85$	Proposed a voxel-coupled convex hull slicing algorithm to calculate individual tree 3D foliage volume by splitting crown point clouds by height and applying distinct volume calculations, enhancing estimation accuracy.	-----	Height-related metrics, density-related metrics, canopy-related metrics
Beyene et al. [52]	Pudu, Selangor, Malaysia	TLS	$R^2 = 0.98$	TLS surpasses optical remote sensing and traditional methods in acquiring tropical forest parameters by offering higher tree detection rates, precise DBH and tree height measurements, and reliable AGB estimations.	Laser wavelengths (1550 nm) and has a measurement ranging from 1.5 to 600 m.	CHM, DBH
Tian et al. [53]	Ejina Banner, Inner Mongolia, China	ICESat, Geoscience Laser Altimeter System	$R^2 = 0.741$	Developed the CHL-BEM biomass estimation model, which integrates vertical canopy structure and horizontal canopy cover information, outperforming traditional models that use only average or maximum tree height.	Laser wavelengths = 1064 nm (near-infrared, NIR), 532 nm (green light); pulse repetition rate: 40 Hz	CTH, CC

However, LiDAR also presents certain limitations in its application. Typically, LiDAR data acquisition is conducted via airborne platforms (such as aircraft and unmanned aerial vehicles) or ground-based systems, resulting in relatively limited coverage and posing challenges for large-scale, rapid monitoring of extensive forested areas. Even with space-borne LiDAR platforms, like NASA's GEDI, the spatial resolution remains insufficient for high-precision forest carbon stock estimations. Additionally, LiDAR data are inherently high-precision three-dimensional point clouds, which entail substantial data volumes. The processing and analysis of such data are complex and necessitate specialized software and significant computational resources for tasks including data preprocessing, feature extraction, and modeling. This complexity is particularly pronounced when handling large-scale datasets, making the processing time-consuming and resource-intensive. Consequently, while LiDAR offers unparalleled accuracy and detail in forest structure measurements, its limitations in coverage, spatial resolution, and data processing complexity must be addressed to fully leverage its potential in forest carbon stock estimation.

2.4. Multi-Source Remote Sensing Data

The application of remote sensing data sources—optical, SAR, and LiDAR—has proven effective in forest carbon stock estimation, with each offering distinct advantages. However, each approach also has limitations that can impact the accuracy and comprehen-

siveness of carbon stock models. To overcome these challenges, integrating multi-source remote sensing data has emerged as a promising solution. By combining the complementary strengths of these data types, it provides more robust and reliable carbon stock estimates. The following table compares the three data types, highlighting their suitability for different forest types and the technical considerations involved in carbon estimation. A comparative analysis of optical, SAR, and LiDAR data for forest carbon stock estimation is presented in Table 4.

Table 4. Comparison of optical, SAR, and LiDAR data for forest carbon stock estimation.

Feature	Optical Data	SAR Data	LiDAR Data
Strengths	High spatial resolution, suitable for capturing spectral information and vegetation indices	Effective in all weather conditions, can penetrate clouds, suitable for nighttime	Provides accurate 3D forest structure measurements, including canopy height and density
Limitations	Limited penetration ability, struggles with dense vegetation, affected by atmospheric conditions (e.g., clouds)	Lower spatial resolution, data interpretation is challenging, less effective in dense forests	High data acquisition cost, limited by scanning angles, requires specialized equipment
Sensitivity to vegetation structure	Sensitive to canopy characteristics (e.g., leaf area index, canopy coverage, and biomass)	Sensitive to surface roughness and vegetation structure, including biomass	Sensitive to canopy height, vertical structure, and topography
Applicable forest types	Suitable for open woodlands, low-density forests, grasslands, and agricultural areas (e.g., tropical and temperate forests)	Suitable for dense forests, wetlands, forest edges, humid regions, effective in subtropical and high-latitude forests	Suitable for high-density forests, tropical rainforests, coniferous forests, and regions with complex forest structures
Technical requirements	Moderate, typically requires image processing and atmospheric correction	High, requires radar signal processing, polarization handling, and efficient interpretation techniques	High, requires LiDAR equipment, point cloud processing, and 3D modeling software
Data acquisition cost	Low, many satellite data available for free (e.g., Landsat, Sentinel)	Moderate, mainly relies on satellite or dedicated radar platforms, some data available publicly	High, requires specialized equipment (airborne or satellite LiDAR), high data acquisition and processing costs

Data fusion techniques, such as pixel-level fusion, feature-level fusion, and decision-level fusion, combine multiple remote sensing data sources into a single, integrated model. This approach enhances the ability to estimate forest carbon stocks by leveraging the unique strengths of each data type simultaneously [54]. Pixel-level fusion merges raw image data from different sensors, allowing for the creation of higher-dimensional data that captures more detailed information on forest structure and biomass. Feature-level fusion, on the other hand, integrates specific features or attributes extracted from each dataset, providing a more comprehensive view of forest dynamics. Decision-level fusion synthesizes the results from separate models, producing a final estimation of forest carbon stocks that incorporates the diverse perspectives offered by each dataset. The integration of these data sources allows for a more accurate and robust estimation of carbon stocks by incorporating complementary information on forest structure, vegetation cover, and environmental conditions. For example, while LiDAR data excel at capturing vertical structure and canopy

height, optical data can provide rich spectral information for vegetation classification, and SAR data can offer insights into forest moisture and roughness. The fusion of these datasets creates a more complete representation of forest ecosystems, addressing gaps left by individual sensors. This approach not only improves the precision of biomass and carbon stock estimates but also enhances the adaptability of models to varying forest types and environmental conditions. By synthesizing the strengths of each sensor type, data fusion techniques significantly contribute to more reliable and scalable remote sensing-based assessments of forest carbon stocks.

To systematically investigate the utilization of multi-source remote sensing data in forest carbon stock estimation, an online scientific literature search was conducted in the Web of Science Core Collection using the keywords “forest carbon stocks”, “remote sensing”, and “multi-source data”. These search terms were specifically chosen to identify studies that employed combinations of different remote sensing data sources for mapping forest carbon stocks or biomass. From this search, 31 papers published between 2019 and 2024 were selected [55–85], focusing on the data sources utilized in multi-source remote sensing-based forest carbon stock estimations. The distribution and usage patterns of various remote sensing data types in these studies are illustrated in Figure 3.

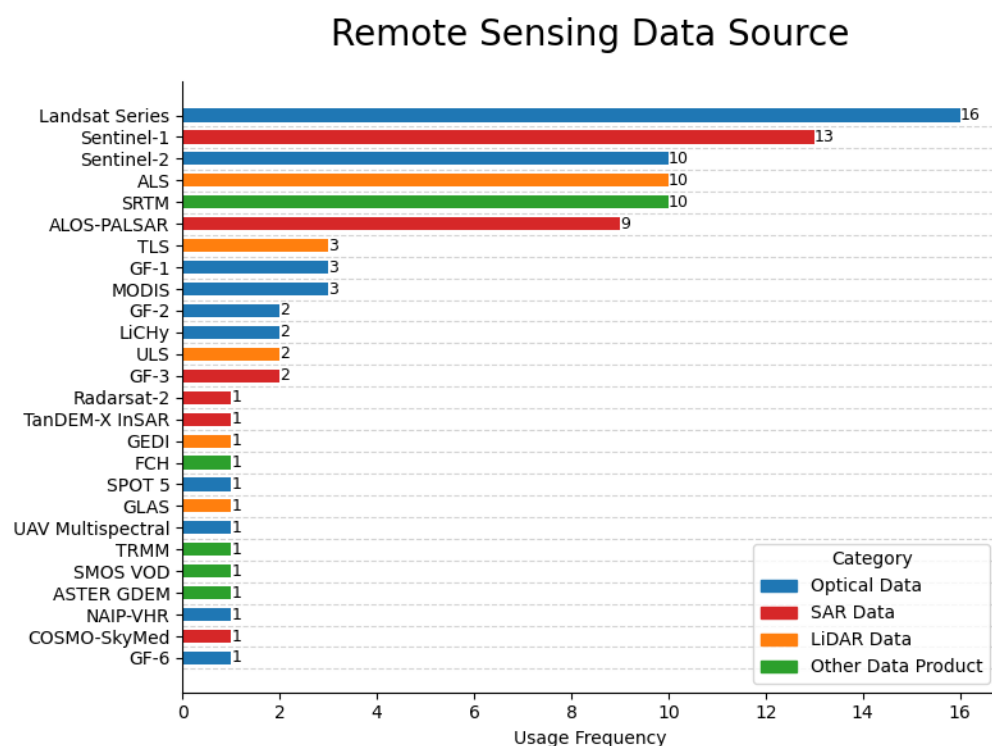


Figure 3. Frequency of remote sensing data sources used in forest carbon stock estimation studies.

According to the statistical analysis of the reviewed literature, among the 31 selected studies, Landsat 8, Sentinel-1, Sentinel-2, airborne laser scanning (ALS), Shuttle Radar Topography Mission (SRTM) and ALOS-PALSAR data sources exhibit high occurrence frequencies. Notably, Landsat series satellite data (such as Landsat 8), Sentinel series satellite data (Sentinel-1 and Sentinel-2), advanced land observing satellite phased-array-type L-band synthetic aperture radar (ALOS-PALSAR), and SRTM data are primarily provided by governmental agencies or related institutions, allowing free access to users and thereby reducing research costs. The maturity of the data processing methods and the availability of supporting software further facilitate widespread utilization among researchers. Additionally, these data sources offer extensive coverage suitable for global or large-area forest carbon stock estimation studies across various scales.

ALS is extensively employed in numerous studies due to its ability to directly acquire three-dimensional spatial information of forests, including tree height, canopy height, and canopy volume. These detailed structural parameters are essential for accurately estimating forest biomass. With the ongoing advancements in remote sensing technology, an increasing number of sensors and data sources have become available. For instance, China's Gaofen series satellites, which encompass multiple satellite types with high spatial, temporal, and spectral resolutions, such as GF-1, GF-2, and GF-3, have been utilized in many studies [86,87]. Additionally, terrestrial laser scanning (TLS), unmanned laser scanning (ULS), and ALS are employed to collect LiDAR data, thereby enhancing the acquisition of comprehensive forest parameters. TLS provides detailed information on lower forest structures, while ALS or ULS offers large-scale forest canopy data [88]. The integration of these three LiDAR technologies enables a more comprehensive description of forest structures and mitigates the limitations associated with lower canopy point cloud deficiencies. This comprehensive approach is suitable for forest areas of varying scales and complexities, meeting the requirements for high-precision estimations in localized regions as well as large-scale forest monitoring. Despite their unique advantages, these new data sources may present challenges related to data acquisition difficulty, higher costs, processing complexity, and lower application maturity, resulting in limited widespread usage. In contrast, the aforementioned commonly used data sources strike an optimal balance between technical feasibility, data quality, and accessibility, thereby dominating current research endeavors.

Among the collected studies on forest carbon stock estimation based on multi-source remote sensing data, the studies summarized in Table 5 have achieved commendable results.

Table 5. Recent studies on forest carbon stock estimation using multi-source remote sensing data.

Study	Study Area	Data Source	Accuracy	Highlight	Fusion Method	Verification
Han et al. [56]	Dabie Mountain Region, China	GF-1, Sentinel-1, SRTM	$R^2 = 0.70$	GF1's NDVI, blue band, and red band texture were significantly correlated with AGB, and elevation was significantly positively correlated with AGB.	Feature-level integration	10-fold cv
Mohite et al. [69]	Forest of India	GEDI, Sentinel-2, Sentinel-1, SRTM	$R^2 = 0.74$	Combined GEDI LiDAR with multi-source satellite and forest soil data to achieve large-scale spatial mapping of forest AGB in India, enhancing forest carbon stock monitoring.	Feature-level integration	5-fold cv
Li et al. [72]	Daxinganling Region, China	Sentinel-1, Sentinel-2, ALOS-PALSAR, SRTM	$R^2 = 0.67$	Compared four data types and their combinations. Found that combining optical and microwave remote sensing with terrain data were most effective for AGB estimation.	Feature-level integration	10-fold cv
Ayushi et al. [73]	Tropical forests of the BRT Tiger Reserve, Indian	Sentinel-1, Sentinel-2, SRTM	$R^2 = 0.82$	Emphasized integrating multi-source auxiliary datasets, providing new approaches for biomass mapping in heterogeneous landscapes.	Feature-level integration	10-fold cv

It is evident that the fusion of multi-source remote sensing data positively enhances the accuracy of forest carbon stock estimations. However, the integration of diverse data types poses several challenges. One of the primary issues is the disparity in spatial and temporal resolution across different data sources, which can lead to misalignment and inconsistencies during fusion [89]. Additionally, the varying data processing techniques and calibration requirements for each data type further complicate the fusion process, leading to potential errors in the final carbon stock estimations. To address these challenges, advanced data preprocessing and normalization methods can align datasets, while machine learning techniques can automate the fusion process. Calibration with ground-truth data and the use of robust fusion algorithms further improve accuracy and reliability in forest carbon stock estimations.

3. Estimation Methods of Forest Carbon Stocks Based on Remote Sensing Technology

The choice of estimation methodology is critical for forest carbon stock estimation, impacting both the accuracy and applicability of models. Estimation methods can be broadly categorized into two types: empirical models and process-based models [90]. Empirical models use statistical relationships between remote sensing data and measured carbon stocks [91], while process-based models simulate biological and physical carbon dynamics within forest ecosystems [92].

3.1. Empirical Models

Empirical models offer an efficient approach for estimating forest carbon stocks, leveraging remote sensing data to establish statistical correlations with biomass and carbon stock measurements [93]. These models are particularly useful for large-scale carbon stock assessments, providing a broad overview of carbon distribution. The key variables used for estimation vary depending on the type of remote sensing data:

- Optical data: Vegetation indices (e.g., NDVI, EVI) and band reflectance are commonly used to estimate biomass [94]. Band combinations and spectral texture features help identify relationships between spectral properties and forest biomass [95].
- SAR data: The backscattering coefficient is a critical variable, providing insights into forest structure and biomass. Polarization modes (e.g., HH, HV, and VV) offer information about different canopy layers and forest density.
- LiDAR data: Key variables, such as canopy height, density, and three-dimensional structural metrics (from LiDAR point clouds) are used to estimate forest biomass and carbon stocks [96]. LiDAR data's ability to capture detailed vertical and horizontal forest structure enhances the accuracy of these estimates [97].

Traditional empirical regression models and machine learning models are two prominent methods used for these estimations. Regression models, including linear, polynomial, and stepwise regressions, assume fixed relationships between variables and are easy to implement, but they may struggle with complex, nonlinear forest dynamics. Machine learning approaches, such as decision trees and neural networks, capture more complex relationships and offer higher accuracy but require large datasets and careful tuning.

3.1.1. Traditional Empirical Regression Model Methods

Traditional empirical regression models—such as linear regression, polynomial regression, ridge regression, LASSO regression, stepwise regression, and geographically weighted regression (GWR)—are widely used for forest carbon stock estimation. These models offer simplicity and efficiency, particularly in environments with extensive ground-truth data.

- Linear regression establishes a direct relationship between dependent and independent variables, but struggles with nonlinear relationships [98].
- Polynomial regression extends linear models to capture nonlinear patterns but may overfit with higher polynomial degrees [99].
- Ridge and LASSO regression introduce regularization to improve model robustness by preventing overfitting [100,101].
- Stepwise regression automates variable selection, improving simplicity and explanatory power [102].
- GWR accounts for spatial heterogeneity by allowing local variations in regression coefficients, enhancing model accuracy in spatially diverse environments [103].

While traditional empirical regression models offer efficiency, they have limitations. They generally assume fixed relationships between variables, which are insufficient for capturing the complex, nonlinear dynamics in forest ecosystems. Additionally, they are sensitive to data quality and outliers, reducing model reliability. The generalizability of these models is also limited, requiring recalibration for different regions or forest types.

3.1.2. Machine Learning Methods

The advancement of computer science has facilitated the emergence of novel methodologies for establishing carbon stock estimation models, with machine learning (ML) standing out as a prominent branch of artificial intelligence dedicated to developing algorithms that autonomously learn and improve from data [104]. To better understand the key differences between traditional empirical regression models and machine learning-based approaches, a comparison of their main features is presented in Table 6.

Table 6. Comparison of traditional empirical regression models and machine learning models for forest carbon stock estimation.

Feature	Traditional Empirical Regression Models	Machine Learning Models
Model assumptions	Assumes fixed relationships between independent and dependent variables, usually linear or simple nonlinear	Does not require explicit assumptions about the relationship between variables, capable of capturing complex nonlinearities
Flexibility and adaptability	Less flexible, inadequate for capturing complex, nonlinear dynamics in forest ecosystems	Highly flexible, adaptable to complex, dynamic data, capable of modeling nonlinear and time-varying patterns
Data requirements	High-quality data required, sensitive to outliers	Lower data quality requirements, can handle missing values and outliers
Computational complexity	Relatively low, simple computation, suitable for small datasets	High computational complexity, especially when dealing with large datasets, may require more computational resources
Applicability	Typically suited for smaller datasets or simpler scenarios, often used for standard forest carbon stock estimation	More suitable for large datasets and complex problems, such as multi-source remote sensing data and varying forest types

Among the various ML algorithms, random forest regression and extreme gradient boosting (XGBoost) are particularly prominent in forest carbon stock estimation.

Random forest regression is an ensemble method that constructs multiple decision trees and aggregates their predictions [105]. This algorithm excels at handling complex, nonlinear relationships and is robust to data noise, missing values, and high-dimensional feature spaces. It is highly generalizable, making it suitable for diverse forest carbon stock

estimation scenarios. Recent advancements, such as geographically weighted random forest regression [106] and Bayesian kriging integrations [107], have further improved its accuracy in carbon stock estimation.

XGBoost, another powerful ML algorithm, is based on ensemble learning and gradient boosting principles [108]. It builds multiple decision trees and combines them through weighted summation, while regularization techniques control model complexity. XGBoost is particularly effective at managing nonlinear relationships between carbon stocks and forest structural features, offering high estimation accuracy and strong potential for forest carbon stock applications [109].

Additionally, other machine learning algorithms have been successfully applied to forest carbon stock estimation, achieving commendable results alongside the previously mentioned random forest and XGBoost algorithms, as summarized in Table 7.

Table 7. Recent studies on forest carbon stock estimation using machine learning methods.

Study	Study Area	Algorithm	Accuracy	Highlight	Verification
Singh et al. [110]	Dhanbad Region, Jharkhand, India	GAMM, K-NN, SVM, ANN, RF	RF: adjusted $R^2 = 0.91$	Integrated machine learning and field surveys enable finer-scale AGB estimation, highlighting new spectral bands' benefits and the importance of uncertainty mapping for enhanced prediction accuracy.	Field data from 106 plots, 5-fold CV
Fararoda et al. [111]	Forests of India	RF	$R^2 = 0.89$	First use of multi-source data and machine learning to estimate aboveground biomass in Indian forests, overcoming single-sensor saturation in high biomass areas and improving accuracy.	Inventory data, 10-fold CV
Qadeer et al. [112]	Diamir District, Gilgit-Baltistan, Pakistan	RF, GTB, CatBoost, LightGBM, XGBoost	LightGBM: $R^2 = 0.734$	Assessed ML algorithms for AGB prediction in complex terrains. LightGBM and XGBoost performed best with specific datasets, handling nonlinearities and improving accuracy.	Field data from 171 plots
Li et al. [113]	Chenzhou, Hunan, China	LR, RF, XGBoost	XGBoost: $R^2 = 0.75$	Examined the importance of parameter optimization in machine learning algorithms.	Inventory data, 10-fold CV

Despite the proven potential of ML methods in forest carbon stock estimation, their application is often limited to specific regions and forest types [114]. A major challenge is the lack of calibrated data across different environmental conditions, which hinders the transferability of models. This makes it difficult to apply models trained in one region to others with different ecological contexts. Additionally, ML models face issues, such as overfitting, where models become overly complex and fit data noise rather than underlying trends. Data bias, arising from limited or unrepresentative training datasets, can further degrade model performance in diverse ecological settings. Moreover, the high computational costs of training sophisticated ML models, especially with large-scale, high-resolution datasets, can pose barriers. To improve model generalization, future research should focus on developing more versatile ML models that can better adapt to varying environmental

conditions, address data calibration challenges, and mitigate overfitting and bias [115]. These improvements would enhance the scalability and applicability of ML models for global forest carbon stock estimation.

3.1.3. Limitations of Empirical Models

Empirical models, including traditional regression models and machine learning approaches, are effective in directly estimating forest carbon stocks by capturing statistical relationships between explanatory variables and observed carbon stocks from sample plots. These models can be used to estimate carbon stocks in unmeasured areas, often providing satisfactory fitting accuracy. However, their performance is limited by the inherent challenges of remote sensing data, such as climatic conditions, sensor calibration, and viewing angles. Issues, like cloud cover, atmospheric disturbances, and image noise, can degrade data quality, affecting the precision of model inversions. Additionally, direct inversion models, which rely on statistical correlations, lack a mechanistic understanding of the biophysical processes driving carbon cycling. This limits their ability to explain the factors influencing carbon stock changes or predict future trends accurately. The applicability of these models is also constrained by the availability of observational data, making them less effective in regions or time periods outside the scope of the training datasets. Consequently, the extrapolation performance of these models decreases when applied to different forest types or regions with varying climatic conditions, limiting their generalizability and predictive accuracy.

3.2. Estimation of Forest Carbon Stocks Using Process Models

While direct estimation methods, such as field-based measurements of DBH, height, and stand density, are straightforward, they are often limited by spatial heterogeneity and measurement errors. Furthermore, these methods struggle to capture the dynamic ecological processes that govern forest ecosystems.

To address these limitations, process-based models have emerged. Unlike empirical models, process-based models simulate carbon cycling within forest ecosystems, considering interactions between climate, soil, and forest management practices [116]. These models offer more detailed and long-term estimates of carbon stocks, making them particularly useful for large-scale regions or areas with sparse field data.

3.2.1. Principles of Process Models

As understanding of biophysical and chemical processes evolves, studies increasingly focus on key factors, such as solar radiation, photosynthesis, respiration, and nutrient cycling [117]. Parameters, like leaf area index, canopy coverage, photosynthetically active radiation, and chlorophyll content, are integrated into mechanistic models to simulate carbon dynamics [118]. Radiation transfer models, which simulate how solar radiation is absorbed by the canopy, are crucial for predicting carbon fluxes. These models provide critical insights into forest productivity and carbon sequestration, though they represent only one aspect of broader process-based models, which incorporate various ecological and environmental factors.

3.2.2. Common Process Models Applied in Forest Carbon Stock Estimation

Several widely used process-based models are employed in forest carbon stock estimation, including the Biome-BGC (biogeochemical cycles), CASA (Carnegie–Ames–Stanford approach), DGVM (dynamic global vegetation model), and CLM (community land model). Each model has its strengths, limitations, and unique features suited for different research scenarios, as follows:

- Biome-BGC: Simulates carbon, nitrogen, and water cycles, providing a detailed model for carbon storage in forest ecosystems [119]. It is especially useful for modeling carbon dynamics under extreme environmental conditions, like drought or excess moisture [120].
- CASA: Focuses on vegetation photosynthesis and estimates net primary productivity from remote sensing data [121]. It has shown good performance in large-scale carbon stock estimation, incorporating biomass density and carbon stock calculations [122].
- DGVM: Models dynamic interactions between vegetation and the environment, simulating processes, like photosynthesis, respiration, and carbon transfer [123]. It is particularly effective for predicting forest carbon stock changes due to climate change [124].
- CLM: A terrestrial ecosystem model that simulates energy, carbon, and water cycling [125]. It provides essential insights for climate change studies and forest carbon modeling at regional and global scales, supporting forest management and policy decisions [126].

In addition to these models, various other process-based models have been applied to forest carbon stock estimation. A summary of recent research using process models for forest carbon stock estimation is presented in Table 8.

Table 8. Recent studies on forest carbon stock estimation using process models.

Study	Study Area	Model	Highlight	Input Parameters	Scale of Study Area
Gao et al. [127]	Global vegetation	IAP-DVGM	Simulated global vegetation and carbon flux responses to warming. Analysis indicates that carbon flux is influenced by leaf area index, temperature, and precipitation.	Meteorological data, vegetation data	Global
Srinet et al. [128]	Himalayan Foothills, India	Biome-BGC	Studied climate impacts on carbon flux and showed forest carbon sink potential.	Meteorological data, site characteristics, ecophysiological parameters	Regional
Bai et al. [129]	Mengjiagang Forest Farm, Heilongjiang Province, China	3-PG	Used a 3-PG model and remote sensing to predict larch DBH and biomass, generating carbon storage maps for artificial plantations.	Meteorological data, soil data, vegetation data	Regional
Wang et al. [130]	Shenzhen, China	CASA	CASA outperforms InVEST spatially. From 2008–2022, Shenzhen’s construction land increased, green and other land uses decreased, and green carbon storage declined.	Meteorological data, NDVI, SR, vegetation data	Regional

3.2.3. Limitations of Process Models

Estimating forest carbon stocks with process-based models offers a dynamic method to simulate carbon changes, incorporating factors, such as climate, environmental conditions, and management practices. These models can adapt to different spatial and temporal scales, providing valuable insights for decision-making and scenario prediction. However, they

require large datasets, including meteorological variables, soil properties, and vegetation parameters, which may be difficult to obtain or lack accuracy. Data quality issues can significantly impact model reliability.

Additionally, the complex ecological processes these models simulate, such as soil carbon release and root carbon input, are challenging to measure directly, complicating model validation. While process models are useful for predicting the impacts of climate and land-use changes on carbon stocks, they are subject to high uncertainty, especially when forecasting future scenarios. Uncertainties in climate projections, vegetation responses, and other variables can significantly increase error margins in predictions.

3.3. Advancing Empirical and Process Models: Comparison, Uncertainty, and the Role of Artificial Intelligence

To fully appreciate the strengths and limitations of empirical and process-based models, it is important to consider their sensitivity to different forest types, the uncertainties inherent in each approach, and how artificial intelligence is increasingly shaping their development.

3.3.1. Comparison of Empirical and Process Models

The suitability of forest carbon stock estimation models varies depending on the forest type, as each model's strengths and limitations align differently with forest structure and environmental conditions. Both empirical and process models have their advantages and challenges when applied to different forest ecosystems, making the choice of model crucial for accurate carbon stock estimation. Table 9 summarizes the sensitivity of each model type to different forest types, as well as their respective strengths, limitations, and the data required for each approach.

Table 9. Comparison of empirical and process-based models for forest carbon stock estimation.

Model Type	Required Data	Strengths	Limitations	Forest Type Sensitivity
Empirical models	- Remote sensing data (e.g., optical, SAR, LiDAR); - Ground-truth data (e.g., biomass samples, field surveys).	High efficiency and simplicity in biomass and carbon stock estimations.	Limited adaptability to complex and heterogeneous forest ecosystems.	- Sensitive to forest structure and spectral characteristics; - Performance may vary across different forest types due to spectral homogeneity.
Process models	- Remote sensing data; - Meteorological data (e.g., temperature, precipitation); - Soil and vegetation data (e.g., nutrient content, leaf area index).	Provide a more holistic, dynamic understanding of carbon cycling.	Require more detailed input data and complex simulations, which may not always be available.	- Better suited for forests with diverse ecological conditions (e.g., tropical, boreal); - More sensitive to environmental variables (e.g., climate, soil, nutrient cycles) compared to empirical models.

Empirical models tend to perform well in relatively homogeneous forest types, such as temperate forests or plantation forests, where spectral characteristics remain consistent and can be easily linked to carbon stocks [131]. These models are advantageous in terms of speed and efficiency, especially in regions with extensive ground-truth data. However, their applicability may be limited in forests with high ecological complexity, where spectral signals are more variable.

Conversely, process-based models are particularly effective in forests with diverse ecological conditions, such as tropical rainforests or boreal forests, where there are significant variations in species composition, canopy structure, and environmental factors [132]. These models take into account various biophysical processes and climate interactions, making them better suited for capturing the dynamic nature of carbon cycling in complex forest environments. However, their reliance on extensive data and complex simulations can make them more resource-intensive compared to empirical models.

In summary, while empirical models offer simplicity and efficiency, they may fall short in complex forest ecosystems, where process-based models excel. The choice between these two approaches should be guided by the forest type, data availability, and the level of detail required for the study.

3.3.2. Model Validation in Empirical and Process Models

Model validation is essential for ensuring the accuracy and reliability of forest carbon stock estimations [133]. In empirical models, validation typically involves comparing predictions with ground-truth data, such as biomass samples or field surveys. Cross-validation techniques are often used to assess the model's generalizability by splitting the dataset into training and validation subsets, helping to avoid overfitting. For process-based models, validation requires long-term field data to assess the accuracy of carbon dynamics simulations, ensuring they reflect real-world processes, like photosynthesis and nutrient cycling.

However, validation faces challenges, particularly in regions with limited ground-truth data. In areas with complex ecological conditions, such as tropical or boreal forests, the scarcity of detailed data on forest structure and climate factors can hinder validation efforts. Furthermore, models trained in one region may perform poorly when transferred to another, underscoring the need for careful calibration to specific regions.

To improve model validation, future research should focus on obtaining more region-specific ground-truth data, particularly in understudied areas. Advanced technologies, like UAVs and high-resolution satellite imagery, can assist in overcoming data gaps. Additionally, employing multi-source data and integrating uncertainty analysis will enhance validation, improving the accuracy and applicability of forest carbon stock estimations across diverse ecosystems.

3.3.3. Uncertainty in Empirical and Process Models

Uncertainty is a key challenge in forest carbon stock estimation, impacting both empirical and process-based models.

In empirical models, uncertainty mainly arises from the quality and availability of ground-truth data [134]. The accuracy of these models depends on how well the calibration data represent the studied area. In regions with limited field data, carbon stock estimates may be less reliable, as empirical models often assume fixed relationships between input variables, which may not apply across different forest types or environmental conditions.

Process models also face uncertainty, particularly due to their reliance on numerous input variables, such as climate data, soil characteristics, and vegetation parameters [135]. The accuracy of these models depends on the precision and availability of these factors, and gaps in data, such as unreliable or missing meteorological information, can lead to errors. The complexity of these models and their numerous interacting processes also introduce additional uncertainties, especially when simulating long-term carbon dynamics under varying environmental conditions.

Both model types are affected by uncertainty, but the nature of these uncertainties differs. While empirical models struggle with data representativeness, process-based mod-

els face challenges in the availability and precision of ecological data. To reduce these uncertainties, improvements in data quality and advancements in modeling techniques are essential. Integrating multiple data sources and refining parameterization can enhance the reliability of carbon stock estimations. Overall, empirical models are limited by gaps in ground-truth data, while process models rely on complex, often incomplete ecological data. Overcoming these challenges requires enhanced data collection and improved modeling methods.

3.3.4. The Growing Importance of Artificial Intelligence in Both Model Types

Artificial intelligence (AI) is playing an increasingly important role in enhancing forest carbon stock estimation methods [136], particularly in empirical and process-based models. In empirical models, machine learning techniques, like random forest approaches and deep learning, are improving predictive accuracy by capturing complex, nonlinear relationships in large datasets. AI allows these models to adapt better to diverse forest types and environmental conditions, boosting reliability and precision.

For process-based models, AI can optimize parameterization and simulate ecological processes with higher accuracy [137]. AI algorithms can help refine complex models by learning from observed data, reducing computational costs, and improving simulations of carbon dynamics under varying environmental conditions.

AI's integration into both model types significantly enhances scalability, adaptability, and accuracy, offering new opportunities for better forest carbon stock estimations. As AI advances, it will continue to improve the precision and efficiency of these models, supporting forest management and climate change mitigation strategies. In conclusion, AI is revolutionizing both empirical and process-based models by improving their predictive capabilities, reducing uncertainty, and making them more applicable across different forest types and conditions.

4. Technological Innovation and the Latest Research Progress

Technological advancements are revolutionizing the use of remote sensing for forest carbon stock estimation. The integration of deep learning techniques has significantly improved both the efficiency and accuracy of carbon stock estimations, particularly in processing large-scale datasets and enhancing pattern recognition capabilities [138]. Research on vegetation optical depth (VOD) has also introduced new ways to capture forest structural characteristics, refining the carbon stock estimation process [139]. Moreover, growing recognition of the bidirectional relationship between forests and climate change is enhancing methods for forecasting future carbon dynamics and informing climate change mitigation strategies [140]. Collectively, these innovations are broadening the applications of remote sensing in forest carbon stock estimation.

4.1. Deep Learning

Deep learning, a subset of machine learning based on artificial neural networks, has shown promise for forest carbon stock estimation, especially in processing complex data, like remote sensing images and time-series data [141]. It excels at capturing intricate patterns in large datasets [142], offering a solution where traditional methods may fall short.

4.1.1. Introduction of Related Algorithms

Deep learning encompasses various classic neural network architectures, each with distinct structures and functions tailored to different types of data and tasks [143]. Among these, two deep learning architectures are particularly well-suited for estimating forest carbon stocks due to their ability to handle complex and high-dimensional data, such as remote sensing imagery and time-series observations.

- Convolutional Neural Networks (CNNs)
- CNNs are particularly effective for processing high-resolution optical images [144]. By automatically extracting features, like tree edges and crown shapes, CNNs enhance the accuracy of carbon stock estimations across various forest types [145]. Their ability to process large-scale, high-resolution images makes them adaptable for diverse environmental conditions and vital for forest management and carbon cycle research [146].
- Recurrent Neural Networks (RNNs) and Long Short-Term Memory Networks (LSTMs).

RNNs and LSTMs excel at handling time-series data, such as monitoring tree growth and climate variations [147]. These models help predict dynamic changes in forest carbon stocks over time by capturing temporal dependencies [148]. LSTMs address issues in traditional RNNs, like the vanishing gradient problem, by improving memory retention [149], which is crucial for modeling long-term climate influences on carbon stocks.

4.1.2. Applications and Limitations of Deep Learning in the Field of Forest Carbon Stocks

Deep learning has become a powerful tool for estimating forest carbon stocks, providing innovative approaches to process large and complex datasets, such as remote sensing imagery and time-series data [150]. By leveraging its ability to automatically learn intricate features, deep learning models can address the complexities of forest carbon estimation that traditional methods struggle with.

In terms of applications, deep learning techniques, like CNNs and RNNs, are particularly effective in specific scenarios. For instance, in tropical reforestation projects, a deep learning-based end-to-end model utilizing individual tree detection from low-cost RGB-only drone imagery has been shown to accurately estimate forest carbon stock within official carbon offsetting certification standards [151]. Additionally, deep learning models have been applied to estimate forest biomass from airborne LiDAR point clouds, providing more accurate estimates compared to traditional methods [152]. These models are also well-suited for large-scale forest carbon monitoring, where they can efficiently process vast datasets, providing timely and accurate carbon stock estimates that traditional methods cannot easily achieve.

However, deep learning models come with several limitations. One major challenge is the high demand for large, high-quality labeled datasets, which are often difficult and expensive to obtain, especially in remote regions. Additionally, these models are highly dependent on the quality and representativeness of training data, and poor generalization to new or diverse forest types can lead to unreliable estimates. Another limitation is the lack of interpretability—deep learning models are often seen as “black boxes”, making it difficult to understand how predictions are made, which poses challenges for validation and practical application in forest management. Lastly, deep learning models require significant computational resources, making them less accessible in areas with limited access to high-performance computing infrastructure [153].

In conclusion, while deep learning offers considerable potential to enhance forest carbon stock estimation, addressing the challenges of data availability, model transparency, and computational demands is crucial for fully realizing its benefits. Continued research into improving model generalization and reducing data dependency will help unlock the full potential of deep learning in forest carbon monitoring.

4.2. VOD

VOD is a critical remote sensing indicator used in forest carbon stock estimation, providing valuable insights into vegetation biomass, water content, and structural charac-

teristics [154]. VOD captures the interaction between vegetation and microwave radiation, making it a crucial tool for forest carbon monitoring and climate change assessment. It enhances the accuracy of carbon stock estimates by offering a clearer understanding of forest biomass and moisture content [155], both of which are tightly linked to carbon sequestration processes.

4.2.1. Acquisition Methods

VOD is derived from microwave sensors operating at various frequencies, such as the SMOS satellite in the L-band and AMSR-E in the C-band and X-band. These sensors enable large-area VOD estimation, which is vital for monitoring forest ecosystems. The inversion process typically utilizes the 0-order Tau-Omega radiative transfer model, which simulates the brightness temperature of the vegetation layer. To improve accuracy, environmental parameters, like soil temperature, surface roughness, and effective scattering albedo are considered, as they significantly impact the VOD estimation process.

4.2.2. Applications of VOD in Forest Monitoring

VOD has proven to be especially useful in the following specific forest environments, where its unique advantages become evident:

- Tropical forests: VOD correlates strongly with forest biomass, particularly in high-biomass areas, offering a more reliable estimate than optical indices. In some regions, like the Amazon and African tropical forests, L-band VOD has been validated as a key tool for biomass estimation.
- Boreal forests: In snow-covered areas where optical data are often compromised, VOD provides consistent measures of forest moisture, enabling year-round monitoring of forest health.
- Dryland ecosystems: VOD also serves as an indicator of drought stress and recovery, tracking changes in vegetation moisture to assess the impacts of water scarcity on forest ecosystems.
- Key advancements include the following:
- Relationship with forest biomass: VOD, particularly at L-band, shows a strong linear relationship with AGB, making it an effective tool for biomass estimation in regions with high forest density [156].
- Monitoring forest ecosystem functions: VOD is closely related to ecological processes, like photosynthesis and water-use efficiency [157]. Studies have demonstrated that VOD can be linked with metrics, such as primary productivity and light-use efficiency, enhancing the understanding of forest ecosystems' response to climate change.
- Forest disturbance and restoration: VOD enables the detection of forest disturbances (e.g., fires, droughts) and the tracking of restoration efforts [158]. It provides real-time updates on vegetation recovery, making it invaluable for ecological restoration practices and forest management.
- Despite its promise, challenges remain with VOD usage. The complexity of inversion algorithms and varying results from different methods highlight the need for optimization and standardization. Additionally, VOD is influenced by various factors, like soil moisture, vegetation type, and topography, complicating its interpretation. To improve the reliability of VOD, future research should focus on refining algorithms, integrating multi-source data, and incorporating machine learning techniques to enhance its utility in forest carbon stock monitoring.

4.3. Bidirectional Influences Between Climate and Forest Carbon Stocks

The relationship between forest carbon stocks and climate is a dynamic feedback loop [159], where forests regulate atmospheric CO₂ concentrations through carbon sequestration, while climate change influences forest health, structure, and carbon storage capacity. Understanding these intricate interactions is crucial for developing effective strategies to mitigate climate change and enhance forest-based carbon sequestration.

4.3.1. The Impact of Climate on Changes in Forest Carbon Stocks

Climate factors, especially temperature and precipitation [160], are key drivers of forest carbon dynamics. Favorable conditions, such as warm and moist climates, enhance photosynthesis and forest productivity, enabling forests to absorb more CO₂ and increase carbon storage.

Conversely, extreme climate events—such as drought, wildfires, and storms—can drastically reduce forest carbon stocks by causing tree mortality, disrupting photosynthesis, and damaging forest structure [161]. Over time, climate change is leading to shifts in species distribution, further impacting carbon sequestration potential [162]. These changes highlight the need for models that account for both current and future climate impacts on forest carbon stocks.

4.3.2. The Impact of Forest Carbon Stocks on Changes in Climate

Forests are essential for regulating the climate through carbon sequestration. By absorbing CO₂ and storing it in vegetation and soil, forests help mitigate global warming [163]. This “climate regulation” role is vital for reducing atmospheric CO₂ concentrations, thereby stabilizing climate systems. Forests also provide vital ecosystem services, including water conservation and soil protection, contributing to ecological balance and climate stability [164]. Their ability to influence surface properties (e.g., albedo and roughness) and the water cycle further supports regional climate regulation, affecting water availability and precipitation patterns.

Forest vegetation plays a critical role in regulating the climate through its structure and function [165]. Vegetation structure influences surface properties, like albedo and roughness. Dense forests increase surface roughness, reduce solar radiation reflection, and raise surface temperatures, while sparse forests have the opposite effect, enhancing radiation reflection and lowering temperatures. This temperature regulation is crucial in regional climate models. In addition, forests influence the water cycle by releasing water into the atmosphere through transpiration, increasing atmospheric water vapor. This not only promotes precipitation but also helps regulate its distribution and intensity. Forests play a key role in redistributing precipitation, which impacts river runoff, groundwater reserves, and regional climate patterns [166].

In conclusion, the relationship between forest carbon stocks and climate is bidirectional, with each influencing the other. Understanding and preserving forest ecosystems is essential for accurate carbon stock estimation and effective climate change mitigation.

5. Discussion

5.1. Challenges

The estimation of forest carbon stocks encounters a range of challenges in both scientific research and practical applications. These challenges are primarily related to data acquisition, model development, forest stand characteristics, and the uncertainties inherent in the estimation process.

5.1.1. Data and Models

Accurately estimating forest carbon stocks using remote sensing data presents several challenges, stemming from both the characteristics of the data and the methods used for analysis. One primary issue is signal saturation, particularly in dense vegetation areas, which can lead to underestimations of carbon stock. For optical data, atmospheric disturbances, such as cloud cover and varying sunlight, can degrade image quality. Radar data are less affected by weather but can still suffer from signal degradation due to rain or surface interference, reducing reliability.

To address these challenges, data fusion techniques that combine optical, radar, and LiDAR data have proven effective. These approaches compensate for the limitations of each data source—radar provides structural data unaffected by weather, optical data offer spectral information, and LiDAR captures vertical structure with high precision. Combining these data types enhances the overall accuracy of carbon stock estimates. Moreover, atmospheric correction and time-series analysis can further improve data reliability across varying conditions.

Another significant challenge is the temporal and spatial resolution of remote sensing data. In complex ecosystems, fine-scale variations in forest structure can greatly influence carbon storage, yet satellite data may not have the spatial resolution needed to capture these changes. High-resolution LiDAR and UAV imagery can help address this challenge by providing detailed structural data. Time-series analysis of multi-temporal satellite data can also track carbon stock changes over time.

Despite advances in data fusion and high-resolution technologies, data gaps and a lack of field validation data remain obstacles. These gaps can be filled using gap-filling algorithms and global datasets that complement local observations. Additionally, machine learning techniques can enhance data continuity and optimize the gap-filling process.

Regarding estimation methods, empirical models, including traditional regression and machine learning approaches, have been widely used to establish statistical relationships between remote sensing data and measured carbon stocks. While these models are efficient, they are limited by data resolution, processing techniques, and the availability of reliable ground data. Moreover, they may lack ecological interpretability, reducing their generalizability to other forest types or environmental conditions.

In contrast, process-based models, which simulate carbon dynamics through ecological parameters, such as photosynthesis, respiration, and nutrient cycling, provide a more comprehensive understanding of carbon sequestration. However, these models require large amounts of high-quality data and significant computational resources. Their accuracy is compromised if input data are insufficient or inaccurate.

Ultimately, the choice between empirical and process-based models depends on the research objectives and data availability. Hybrid models, combining the efficiency of empirical methods with the ecological depth of process-based models, offer a more reliable framework for forest carbon stock estimation across different ecosystems.

In conclusion, while challenges in data and modeling exist, they can be overcome through the integration of multiple data sources, advanced processing techniques, and hybrid models, enhancing the precision and applicability of forest carbon stock estimations.

5.1.2. Forest Stand

The intricate structure of global forest stands presents numerous formidable challenges to the accurate estimation of forest carbon stocks, primarily manifested in the following aspects.

Firstly, the diversity in stand structures results in a wide variety of forest types. Distinct forest ecosystems, such as tropical rainforests, temperate broadleaf forests, and

boreal coniferous forests, exhibit significant variations in tree species composition, age distribution, and stand architecture [167]. These differences cause variations in signal reflection, such as differences in how various tree species or forest layers reflect light, making it difficult to develop universal carbon stock estimation models. For instance, certain tree species may have different spectral signatures due to leaf structure or canopy density, which complicates the use of remote sensing data for biomass estimation. This heterogeneity not only complicates model development but also reduces the accuracy and generalizability of the estimations.

Secondly, significant intraspecific variability exists among individual trees within the same stand [168]. Trees within a given stand vary in their morphological characteristics, such as height, diameter at breast height (DBH), and crown width, as well as in their growth rates. This variability increases the complexity of assessing forest biomass and carbon stocks. In particular, some methods, like averaging or using simple statistical models, may not accurately represent the diversity in tree structures. Machine learning approaches, such as random forest or neural networks, may improve this by capturing complex relationships between tree characteristics and carbon storage, but their sensitivity to tree crown closure and vertical stratification changes must be carefully validated.

Moreover, the vertical stratification of stands is complex, comprising multiple layers, such as the canopy, shrub, and herb layers. The interactions among these layers and their respective contributions to carbon storage are difficult to quantify precisely [169]. For example, estimating the contribution of understory vegetation or shrubs to the overall carbon stock is challenging due to their small size and limited biomass relative to the overstory. Multi-source data fusion, combining optical, radar, and LiDAR data, can help estimate biomass across these different layers more effectively. However, integrating these data sources requires advanced algorithms and robust calibration methods to ensure accurate and consistent results.

Additionally, the spatial distribution of forests is often heterogeneous, with such features as forest gaps and edges [170]. Carbon stocks and ecological processes in these regions can differ markedly from those within the interior of the stand. Accurately capturing this spatial heterogeneity, particularly in large-scale estimations, requires scaling models that can account for micro-environments within forest stands. Process-based models and data assimilation techniques have shown promise in capturing this spatial variability, yet their scalability and applicability to large, heterogeneous forest complexes remain a key challenge.

Finally, forest ecosystems are subject to various natural and anthropogenic disturbances, including fires, pests, diseases, and logging activities [171]. These disturbances alter stand structure, impede forest growth, and disrupt carbon cycling processes, adding another layer of complexity to carbon stock estimation. Addressing these challenges requires dynamic models that account for disturbance impacts, such as the effect of pest infestations on carbon dynamics or the recovery of carbon stocks after forest fires. The integration of disturbance models with remote sensing data and long-term field observation can enhance the accuracy of carbon estimates, but the lack of widely accepted scalable procedures for these dynamic adjustments remains a significant gap.

In conclusion, the complex structure of global stands imposes numerous severe challenges on the estimation of forest carbon stocks. However, solutions to overcome these challenges are emerging. Advanced methods, such as machine learning algorithms, multi-source data fusion, and process-based models, offer promising pathways for improving the accuracy and reliability of estimations. By combining long-term field observation data with these innovative modeling techniques, future efforts can address the complexities of forest heterogeneity, ultimately enhancing the precision of carbon stock estimates. Fur-

ther research into scaling and applying these models to larger, more heterogeneous forest complexes is critical for advancing forest carbon monitoring and management.

5.1.3. Uncertainty

The estimation of forest carbon stocks is pivotal for elucidating the role of forest ecosystems within the global carbon cycle; however, numerous uncertainties inherent in the estimation process significantly challenge the accuracy of these assessments.

One primary source of uncertainty arises from the characteristics of remote sensing data [172]. For instance, utilizing remote sensing imagery of varying resolutions—such as National Agriculture Imagery Program images and LiDAR point clouds—introduces variability in resolution and point cloud density, which can substantially influence estimation outcomes. The variability in these datasets necessitates careful consideration of calibration and validation techniques, particularly in multi-source datasets. Calibration procedures, such as adjusting for sensor differences, are essential to mitigate inconsistencies across different sensors, like Landsat OLI and Landsat ETM+, while validation processes, particularly cross-validation, can help assess model performance under diverse conditions, improving reliability. Additionally, the preprocessing of data from disparate sensors, including radiometric calibration, atmospheric correction, and topographic correction, introduces potential sources of error. Identifying and standardizing preprocessing methods are crucial to minimizing uncertainties across datasets and ensuring harmonization.

Furthermore, the selection of sample plots is often subject to subjective biases, and the measurement of tree parameters—such as tree height and DBH—is prone to errors stemming from measurement tools and methodologies [173]. These errors can be reduced by applying robust statistical validation methods, such as bootstrapping or Bayesian approaches, to quantify and correct for biases in field measurements and sample selection. Human-induced disturbances, including logging, grazing, and tourism, vary across regions and forest types, adding layers of complexity and uncertainty. Quantifying these disturbances, particularly their temporal variability, can be improved through uncertainty propagation methods, such as Monte Carlo simulations, which incorporate disturbance effects into uncertainty models. This will enable more accurate predictions of their impact on forest structure and carbon stocks.

Spatial scale conversion also introduces uncertainty; transitioning from local to regional or global scales alters the relationship between forest structure and carbon stocks, resulting in discrepancies in estimation accuracy as pixel size increases [174]. This scale effect complicates the direct comparison and integration of estimation results across different studies and applications. To reduce scale-related uncertainty, multi-scale modeling approaches and hierarchical Bayesian models can be employed to account for scale effects and provide more accurate results when aggregating data at various spatial resolutions. These methods allow for the incorporation of varying levels of uncertainty depending on the spatial scale of analysis, improving both the consistency and applicability of carbon stock estimates.

Collectively, these intertwined uncertainty factors impede the determination of optimal remote sensing data resolutions and preprocessing techniques, hinder the development of unified estimation models, and elevate the complexity and difficulty of carbon stock estimations. However, addressing these uncertainties through statistical methods, model calibration, and rigorous validation procedures is critical for improving the accuracy and reliability of carbon stock estimates. In the context of climate policy or forest resource management, where accurate data are essential for decision-making, quantifying and controlling uncertainty is indispensable. The application of comprehensive uncertainty analysis

and robust statistical frameworks will enhance the practical utility of these estimates and facilitate their integration into policy and management strategies.

5.2. Future Prospects

The future of forest carbon stock estimation via remote sensing will be driven by advancements in artificial intelligence, multi-source data integration, and methodological standardization. This section outlines both short-term implementable solutions and long-term research directions, emphasizing practical adoption and scalability.

5.2.1. Short-Term Implementable Solutions

AI-driven automation for carbon stock estimation: AI enhances efficiency and accuracy by automating feature extraction and classification [175]. Deep learning models, such as convolutional neural networks and transformers, improve high-resolution image processing. Future efforts should focus on transfer learning for model adaptation across forest types and self-supervised learning to reduce reliance on labeled training data [176].

Improved multi-source data fusion and calibration: Integrating optical, SAR, LiDAR, and microwave-based data improves estimation accuracy but requires addressing discrepancies in spatial resolution, acquisition timing, and sensor characteristics [177]. Standardized correction algorithms, refined data assimilation techniques, and advancements in VOD-based biomass estimation will help overcome these challenges.

Pilot projects for AI-driven multi-source remote sensing: Pilot projects should target data-rich forest regions, combining multi-source remote sensing with machine learning to develop tailored carbon stock estimation models. Once validated, these models can be scaled to diverse forest ecosystems, providing a structured roadmap for AI-driven forest carbon assessments [175].

Standardization and open data policies for carbon accounting: A lack of standardized methodologies and open-access datasets hinders practical implementation. Immediate efforts should focus on benchmark datasets integrating field inventory and remote sensing data, metadata standardization for seamless integration into carbon reporting systems, and expanding open-access platforms with preprocessed biomass layers [178].

5.2.2. Long-Term Research Directions

AI-augmented process-based models for dynamic carbon accounting: Hybrid AI-process models integrating machine learning with ecosystem simulations will refine carbon flux estimation and improve climate-adaptive assessments [175,179]. Incorporating real-time environmental data will enhance predictive capabilities, enabling better forest response modeling under changing climate conditions.

High-frequency, high-resolution carbon monitoring: Next-generation satellites will provide near-continuous monitoring at unprecedented scales [180]. Expanding LiDAR-based global carbon mapping, leveraging geostationary satellites for sub-daily productivity measurements, and integrating CubeSat and UAV-based monitoring will enable localized, high-resolution assessments.

Climate-responsive carbon stock estimation and policy integration: Long-term carbon stock estimation must account for climate-forest feedback mechanisms. Refining models to assess forest resilience under extreme climate events and aligning remote sensing with policy frameworks will enhance decision-making relevance [181]. Standardized carbon offset indicators will ensure integration into international climate agreements.

The future of remote sensing-based forest carbon stock estimation hinges on AI-driven automation, multi-source data integration, and methodological standardization. Short-term efforts should prioritize data harmonization and algorithm improvements, while long-term research must address climate adaptation, dynamic modeling, and policy integration.

Collaboration between researchers, policymakers, and space agencies is essential to bridge theory and practical implementation, ensuring that remote sensing plays a central role in sustainable forest management and carbon mitigation.

6. Conclusions

Accurate forest carbon stock estimation is crucial for understanding carbon cycles and supporting climate policies. This review emphasizes key advancements in remote sensing-based carbon assessment, particularly in deep learning models, multi-source data integration, and AI-driven large-scale monitoring. Deep learning models, such as CNNs and transformers, enhance estimation accuracy by capturing complex patterns in spatial and spectral data. Multi-source data fusion, integrating optical, SAR, and LiDAR data, improves precision, especially in high-biomass forests. Despite these advancements, further work is needed to refine model adaptability and address sensor harmonization challenges.

Improved forest carbon estimation has significant policy implications, enhancing carbon inventory accuracy and supporting international frameworks, like the Paris Agreement and REDD+. AI-driven monitoring facilitates real-time tracking of forest changes, enabling more responsive management. Expanding AI-assisted databases will increase transparency and scalability in global carbon monitoring efforts. Future research should focus on bridging the gap between high-resolution research and practical implementation. Collaboration among researchers, policymakers, and global organizations is essential to develop robust and actionable carbon estimation frameworks. Through the integration of AI, remote sensing, and standardized protocols, remote sensing will play a pivotal role in advancing climate action.

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Abbreviations

The following abbreviations are used in this manuscript:

GCP	Global Carbon Project
AGB	Aboveground biomass
DBH	Tree diameter at breast height
SAR	Synthetic aperture radar
LiDAR	Light laser detection and ranging
NIR	Near-infrared
GF	China's Gaofen series satellites
NASA	National Aeronautics and Space Administration
GED	Global Ecosystem Dynamics Investigation
ICESat-2	Ice, Cloud, and Land Elevation Satellite-2

SRTM	Shuttle Radar Topography Mission
TLS	Terrestrial LiDAR system
ULS	Unmanned Lidar system
ALS	Airborne LiDAR system
NDVI	Normalized difference vegetation index
EVI	Enhanced vegetation index
LASSO	Least absolute shrinkage and selection operator
GWR	Geographically weighted regression
ML	Machine learning
XGBoost	Extreme gradient boosting
PAR	Photosynthetically active radiation
Biome-BGC	Biogeochemical cycles
CASA	Carnegie–Ames–Stanford approach
NPP	Net primary productivity
DGVM	Dynamic global vegetation model
CLM	Community land model
NCAR	National Center for Atmospheric Research
CNN	Convolutional neural network
RNN	Recurrent neural network
LSTM	Long short-term memory network
VOD	Vegetation optical depth
AI	Artificial intelligence

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